## Finland
## Finnish Competition and Consumer Authority

The Finnish Competition and Consumer Authority (FCCA) is committed to improving its effectiveness through the adoption of sophisticated computational tools. In particular, the authority is exploring methods to employ artificial intelligence, including Large Language Models (LLMs), in its daily operations. The authority is providing its staff with the tools and training to use LLMs effectively. In addition, it deploys computational tools in areas such as cartel detection and the maintenance of competitive neutrality.

A primary objective of the FCCA's competition division is the identification and elimination of cartels, given their significant and detrimental impact on competition. To this end, the FCCA has instituted a two-stage screening process to detect cartels from bidding data, with a specific emphasis on public procurement tenders. This process leverages automated computational tools to optimize efficiency and precision.

The initial stage of the cartel screening process involves the systematic application of various methodologies to a large dataset on public sector tenders, with the aim of identifying potential collusion indicators warranting further scrutiny. This stage is largely automated, employing a wide array of generalizable tests with varying thresholds. As a result of this first stage, certain markets are flagged as potentially suspicious.

The subsequent stage entails a more comprehensive analysis of the markets initially flagged. Since the markets in the database are very heterogeneous, this step of the analysis requires a higher degree of customization and a reduced reliance on automation. The objective is to distinguish between genuinely suspicious markets and those flagged due to false positives. The guiding principle is to ascertain whether the results from the initial stage could be attributed to factors other than collusion.

Upon completion of the second stage, any firms or markets still deemed suspicious are internally reported to the authority's cartel detection function, which determines the subsequent steps of the investigation. The quantitative analysis throughout the screening process is undertaken by the authority's economists, ensuring a robust theoretical and practical foundation.

The FCCA is also responsible for the oversight of competitive neutrality regulation, aimed at ensuring fair competition between the public and private sectors. The principle of competitive neutrality means that public businesses should not have any undue advantages over their private competitors because they are government-owned. The authority is frequently consulted by public sector entities seeking to ensure that they are not infringing the regulations.

In order to be able to respond more efficiently to these queries, the FCCA has piloted a chatbot trained on a dataset comprising previously asked questions and their responses, as well as question-answer data from the authority's web pages. The diversity of the questions necessitates a more complex approach than simple few-shot inference. Yet, the training data is not sufficiently extensive to allow for fine-tuning the model. The approach is currently being tested in a 128k token GPT-4 LLM, which can process all training data and assist in drafting an expert response. Future development plans include a combination model based on Azure Prompt Flow, capable of handling larger datasets with embedding and web search capabilities.

The objective is not to provide advice directly to the inquirer, but to expedite the creation of a response by the authority's staff. The model operates within the Finnish government's private cloud, with no data directly used to train the LLM. While currently applied only to competitive neutrality data, the success of the pilot program could see this setup extended to other customer inquiry response needs.

In addition to the above projects, the FCCA is also leveraging web scraping for a broad spectrum of purposes related to both antitrust cases and policy reports. Web scraping allows for the extraction of large amounts of data from websites quickly and efficiently, and is typically used in cases where alternative data sources are unavailable.